

## NIR-to-RGB Talk — Lectern Cheat-Sheet

Budgeted  $\approx 17:10$  (you ran  $\approx 12$  live). Thread: *judge by recovered behaviour, not image quality*. Set on #4, pay off #11 & #13.

| #  | Slide                 | Say this (one line)                                                                                    | Min  | Cum   |
|----|-----------------------|--------------------------------------------------------------------------------------------------------|------|-------|
| 1  | <b>Title</b>          | Sam Barber — NIR-to-RGB translator so RGB models work on infrared.                                     | 0:15 | 0:15  |
| 2  | <b>Why NIR?</b>       | Hand: NIR penetrates skin (veins), sees in dark. No colour $\rightarrow$ RGB models misread it.        | 0:45 | 1:00  |
| 3  | <b>Problem</b>        | Same scene: RGB works, NIR fails. Retraining every model is impractical.                               | 1:10 | 2:10  |
| 4  | <b>Idea + thesis</b>  | <b>Read the box.</b> Translator in front of a frozen model. Judge by behaviour, not looks.             | 1:10 | 3:20  |
| 5  | <b>Why hard</b>       | <b>(compress if behind)</b> One-to-many; parallax; diffusion too slow/stochastic $\rightarrow$ 2 bets. | 0:50 | 4:10  |
| 6  | <b>Data / rig</b>     | Beam-splitter $\rightarrow$ shared centre $\rightarrow$ one homography. 12,536 pairs, London.          | 1:20 | 5:30  |
| 7  | <b>Aligning pairs</b> | 5-step pipeline: sharpness, crop, SIFT+RANSAC, warp. Optics make one warp work.                        | 0:50 | 6:20  |
| 8  | <b>Core method</b>    | <b>(give it room)</b> Swap VGG for DINOv2+CLIP+ResNet. $F(T(x)) \approx F(\text{RGB})$ .               | 1:50 | 8:10  |
| 9  | <b>Deployable</b>     | Distil 116M $\rightarrow$ 1.7M ( $67\times$ ), mixed-INT8, 2.3 MB. LiteRT + Hailo.                     | 1:00 | 9:10  |
| 10 | <b>Qualitative</b>    | Real outputs. All look plausible — <b>watch Pix2Pix (row 2)</b> . Eye test misleads.                   | 0:55 | 10:05 |
| 11 | <b>Why not PSNR</b>   | Same person/blue shirt 13 dB; postbox 18 dB. <b>Wrong one scores higher.</b>                           | 0:45 | 10:50 |
| 12 | <b>Gap closure</b>    | 0 = no better than NIR, 1 = full RGB recovered. Model's own RGB as reference.                          | 1:05 | 11:55 |
| 13 | <b>Headline</b>       | <b>SLOW + PAUSE.</b> Student +ve on all 5. Pix2Pix ties pixels, NEGATIVE downstream.                   | 1:45 | 13:40 |
| 14 | <b>Ablation</b>       | Drop the ensemble: <b>best PSNR but downstream collapses</b> . Proves it's causal.                     | 1:00 | 14:40 |
| 15 | <b>Edge / demo</b>    | 5.05 FPS CPU, 27 FPS NPU. INT8 faster than FP32. Live app on Pi.                                       | 1:10 | 15:50 |
| 16 | <b>Limitations</b>    | <b>(fast)</b> Behaviour not accuracy; people untested; London-only; sharp frames only.                 | 0:40 | 16:30 |
| 17 | <b>Conclusion</b>     | <b>Name the 3 novelties.</b> Rig/dataset, NAFNet-first + ensemble, edge student.                       | 0:40 | 17:10 |

**Pace check:** at the rig (#6) you should be near **5:30**.  $\approx 17:10$  budget (you ran  $\approx 12$  live). If pressed, drop #7 (aligning) and compress #5, #16.

**Always say** “behaviour agreement”, never “accuracy”. **Pre-load** the demo asset.

### Q&A — question $\rightarrow$ backup slide

- **Circular / pseudo-labels?**  $\rightarrow$  Independent probes (DINOv2-L, SigLIP2) held out, still close  $>0.5$ .
- **DINOv2-S / CLIP closures?**  $\rightarrow$  *Backup: embedding agreement* — higher but training-overlapping, supporting only.
- **Detection weaker than seg?**  $\rightarrow$  Small objects + fine texture reproduced least faithfully.
- **Why NAFNet?**  $\rightarrow$  *Backup: why NAFNet* — activation-free  $\rightarrow$  clean INT8 / edge export.
- **How aligned?**  $\rightarrow$  *Backup: aligned pairs* — beam-splitter removes parallax  $\rightarrow$  one homography.
- **How general?**  $\rightarrow$  *Backup: generalises (and breaks)* — thermal yes, SAR no.
- **Why not diffusion?**  $\rightarrow$  Iterative + stochastic; violates latency + determinism (F3).
- **Why not fine-tune downstream?**  $\rightarrow$  Needs weights + labelled NIR per task; out of scope.

Don't know? Say so, then reason aloud — honesty is marked.